BEFORE YOU READ

Sets by how many elements they hold

A set is finite when its elements can be counted to an end and infinite when they cannot. The universal set is the frame for everything under consideration.

L0 **Types of sets**

- L1 **Finite** a well-defined number of elements — $\{1, 2, 3, 4, 5\}$
  - L2 **Empty set** a special case of a finite set — $\emptyset$
- L1 **Infinite** cannot be exhausted by counting — $\{1, 2, 3, \ldots\}$
- L1 **Universal** all objects under consideration; finite or infinite — U

TAKEAWAY

Size is cardinality

How to compare the sizes of sets, including infinite ones, is the subject of **cardinality**.